

WindOps AI Copilot

Maintenance Action Plans

Scenario: RED

Date: 08 July 2026

Turbines analysed: 3

WTG-12

HIGH

Fault Hypothesis

Severe aerodynamic degradation (pitch misalignment or yaw offset) compounded by instrumentation fault or sensor drift on primary power/SCADA channels

Recommended Action

1) Immediately dispatch technician for a physical pitch angle inspection on all three blades - measure and compare blade root pitch angles against nominal set-point; recalibrate pitch actuators if deviation $>1.5^{\circ}$. 2) Verify yaw position encoder and nacelle vane alignment against wind direction SCADA logs. 3) Inspect and recalibrate the power transducer and anemometer responsible for the 38.3% power gap reading; cross-check against neighbouring turbine wind speed/power curves. 4) Pull anomaly_flag event log and review CMS vibration channels for corroborating signals before returning turbine to full operation.

Rationale

WTG-12 holds the highest fleet priority score (0.896) with 244.9 MWh total energy loss. aero_risk is critically elevated at 0.987, with near-zero mech_risk (0.007), pointing squarely to an aerodynamic root cause rather than drivetrain degradation. The 38.3% power gap confirms sustained underperformance against expected yield. Simultaneously, anomaly_risk is 0.917 with anomaly_flag=1 and anomaly_score=0.791, indicating active instrumentation anomalies that could be masking the true severity - or independently contributing to the power deficit via faulty control-loop feedback. Gear oil temperature (61.6°C) is within normal range, ruling out gearbox involvement at this stage.

WTG-01

HIGH

Fault Hypothesis

Aerodynamic misalignment - likely pitch angle deviation on one or more blades causing asymmetric rotor loading and sustained power deficit

Recommended Action

1) Schedule urgent pitch system inspection: connect to the pitch controller PLC on each blade and read back actual vs. commanded pitch angle; replace or recalibrate any pitch encoder showing deviation $>1^\circ$. 2) Review yaw error statistics from SCADA over the last 30 days - if mean yaw error exceeds $\pm 5^\circ$, perform nacelle alignment correction. 3) Run a power curve verification test (minimum 2-hour window at stable wind speeds 8-12 m/s) post-correction to confirm 35.7% power gap is resolved. 4) Check blade surface condition during the site visit for leading-edge erosion or ice accumulation that could further degrade aerodynamic efficiency.

Rationale

WTG-01 is ranked #2 fleet-wide with priority score 0.887 and 231.1 MWh energy loss. `aero_risk` is 0.969 - near-critical - while `mech_risk` remains negligible at 0.013, making a mechanical drivetrain fault very unlikely. The 35.7% `power_gap_pct` is consistent with pitch misalignment rather than wind resource variability. `anomaly_risk` (0.624) and `anomaly_score` (0.679) are moderately elevated but `anomaly_flag=0`, suggesting the anomaly detection has not yet triggered a hard alert - this pattern is consistent with a gradually developing pitch fault rather than sudden sensor failure. Gear oil temperature (62.2°C) is normal, confirming gearbox is not implicated.

WTG-06

MEDIUM

Fault Hypothesis

Instrumentation fault or sensor drift on power measurement or anemometry channels, with a secondary aerodynamic underperformance contribution

Recommended Action

1) Prioritise sensor audit: inspect and cross-calibrate the nacelle anemometer, power transducer, and wind vane against a calibrated reference or the met-mast closest to WTG-06; replace any sensor showing >5% drift. 2) Review the anomaly_flag=1 event log to identify which SCADA channel(s) triggered the anomaly detection model - isolate the offending signal. 3) Assess pitch system performance via SCADA pitch-angle logs; if pitch tracking error is >1° on any blade, escalate urgency and schedule physical pitch calibration. 4) Investigate the 44.6% power gap (the highest of the top 3) against wind speed data from adjacent turbines - if corrected power gap remains >20% after sensor recalibration, escalate to a full aerodynamic inspection.

Rationale

WTG-06 shows a distinctive pattern of high anomaly_risk (0.650) with anomaly_flag=1 and a high anomaly_score (0.789), combined with only moderate aero_risk (0.557) and very low mech_risk (0.054). The strikingly high power_gap_pct of 44.6% - larger than both higher-ranked turbines - is disproportionate to the moderate risk scores, which is a hallmark of sensor drift inflating or distorting the measured power deficit. Gear oil temperature (53.5°C) is comfortably normal, eliminating drivetrain concerns. Urgency is set to medium given the lower overall priority score (0.477), but the power gap warrants prompt sensor validation before it drives further undetected losses.